

Yuqi Zhao

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Education

PhD	Politecnico di Torino , Control and Computer Engineering • Supervisor: Prof. Marco Mellia • Research Topic: Continuous Machine Learning for Cybersecurity • Research Interests: Network Traffic Analysis, Explainable AI (xAI)	Nov 2024 – Present
MS	Beijing University of Posts and Telecommunications (BUPT) , Computer Science and Technology • GPA: 85/100, Ranking: Top 1% • Supervisor: Prof. Xiaohong Huang • Master's Thesis: A Cluster-Asynchronous Federated Multi-Task Learning	Sep 2020 – Jun 2023
BS	Queen Mary University of London , Engineering • Bachelor of Science with First Class Honours • Joint Bachelor Degree Programme with BUPT BUPT , E-Commerce Engineering with Law • GPA: 86.06/100, Ranking: Top 9% • Supervisor: Prof. Gareth Tyson • Bachelor's Thesis: A Mobile App for Collecting Network Measurement Data	Sep 2016 – Jun 2020

Publications

The Sweet Danger of Sugar: Debunking Representation Learning for Encrypted Traffic Classification Yuqi Zhao , Giovanni Dettori, Matteo Boffa, Luca Vassio, Marco Mellia in Proceedings of the ACM SIGCOMM 2025 Conference (SIGCOMM '25) . Association for Computing Machinery, New York, NY, USA, 296–310.	SIGCOMM 2025
ShortcutCatcher: Making Traffic Classification Reliable Yuqi Zhao , Matteo Boffa, Luca Vassio, Marco Mellia Proceedings of the ACM on Networking, Volume 4, Issue CoNEXT2, Article 23 (June 2026) , 15 pages.	CoNEXT 2026
The Sweet Danger of Sugar: Debunking Representation Learning for Encrypted Traffic Classification with Explainable AI Yuqi Zhao , Giovanni Dettori, Matteo Boffa, Luca Vassio, Marco Mellia Under Major Revision	IEEE/ACM ToN

Projects

Artificial Intelligence for Cyber Threat Intelligence • Focus on sensitive information detection and high-quality samples generation, to reduce overfitting and enhance model generalization to real-world scenarios	May 2025 – Present
Joint Research on IPv6 Network Management: Research Development and Demonstration • Federated Learning System based on Consortium Chain: Implement a reliable and secure federated learning system with Consortium Chain • Active measurement system based on IPv6: Implement an active measurement system by devices probing to assess current IPv4/IPv6 networks.	Sep 2020 – May 2023

Experience

RA **Université d'Angers**, Information

Sep 2023 – Aug 2024

- **Supervisor:** Prof. Jin-Kao Hao **Co-supervisor:** Prof. Eric Monfroy
- **Research Topic:** Hybrid Heuristic Methods for the Multi-Agent Path Finding Problem

Teaching

Machine Learning for Networking, Teaching Assistant

Sep 2025 – Jan 2026

- Designed and delivered hands-on laboratory sessions on network data analysis and supervised learning methods
- Co-designed course projects and mentored student teams on experimental design and technical reporting

Internet Application, Teaching Assistant

Mar 2021 – Jul 2021

- Designed experimental labs for application-layer protocols and Internet systems
- Guided students in hands-on programming exercises and evaluated coursework and projects

Skills

Networking & Systems: Wireshark, Scapy, Network Traffic Collection and Analysis

Machine Learning Methods: Representation Learning, Self-supervised /Pre-training, xAI, Shortcut Learning

Machine Learning Frameworks: PyTorch, AutoGluon, Distributed Data Parallel (DDP)

Optimization & Heuristics: Genetic Algorithms, Local Search

Service

Shadow TPC, ACM CoNEXT 2026

Awards

First Place, [OpenAI Challenge at Politecnico di Torino \(2026\)](#)

Travel Grant, Network Traffic Measurement and Analysis Conference (2026)

Outstanding Graduate Award, Beijing University of Posts and Telecommunications (2023)

Outstanding Undergraduate Award, Beijing University of Posts and Telecommunications (2020)

Academic merit scholarship (2017–2022)